Carcinoma, oncocytic 8290/3
Site Code: Thyroid C73.9

PATIENT HISTORY:

CHIEF COMPLAINT/PRE-OP/POST-OP DIAGNOSIS: Thyroid nodule.
PROCEDURE: Not answered.
SPECIFIC CLINICAL QUESTION: Not answered.
OUTSIDE TISSUE DIAGNOSIS: Not answered.
PRIOR MALIGNANCY: Not answered.
CHEMORADIATION: Not answered.
ORGAN TRANSPLANT: Not answered.
IMMUNOSUPPRESSION: Not answered.
OTHER DISEASES: Not answered.

UUID:84EC1EDD-A480-49F7-905F-8906B7760A94
TCGA-BJ-A192-01A-PR Redacted

SPECIMEN(S) RECEIVED:

1: Right Thyroid Lobe
2: Left Thyroid Lobe and Cervical Thymus

ADDENDUM

MOLECULAR ANATOMIC PATHOLOGY TESTING:

Part 2:

Mutations in *BRAF, NRAS61, HRAS61, KRAS12/13,* and *RET/PTC1, RET/PTC3* and *PAX8/PPARg* rearrangements NOT identified.

NOTE:
Nucleic acids were extracted in the amount sufficient for testing. Amplification of *GAPDH* and *KRT7* controls was acceptable.

BACKGROUND:
Mutations in either *BRAF* or *RAS* genes or *RET/PTC* rearrangements are found in more than 70% of papillary thyroid carcinomas (1). *BRAF* V600E (mutation has been associated with more aggressive behavior of papillary carcinoma (2) and may be used for preoperative risk stratification of patients with papillary thyroid cancer (3). Mutations in the *RAS* genes or *PAX8/PPARg* rearrangement occur in ~70% of follicular thyroid carcinomas and with lower frequency in oncocytic (Hürthle cell) carcinomas (4). Based on a recent prospective study of FNA samples (5) and our experience at UPMC, detection of *BRAF* mutations and *RET/PTC* and *PAX8/PPARg* rearrangements in thyroid FNA samples correlated with ~100% probability of cancer, whereas detection of *RAS* mutation correlated with malignancy in 83-87% of cases and with benign follicular adenoma or hyperplastic nodule in 13-17%. However, single cases of false-positive *BRAF* mutation detection in thyroid FNA samples have been reported in the literature (6) and therefore the results of molecular testing should be interpreted in combination with clinical information, imaging, and cytology.

FINAL DIAGNOSIS:

PART 1: THYROID, RIGHT LOBE, LOBECTOMY (5 GRAMS) –
SLIGHT NODULAR THYROID HYPERPLASIA.

PART 2: THYROID AND THYMUS, LEFT "MASS", COMPLETION THYROIDECTOMY AND THYMECTOMY (42 GRAMS) –

A. **PAPILLARY THYROID CARCINOMA, CYSTIC, ONCOCYTIC VARIANT (5.0 CM), CONFINED TO THE THYROID; NO ANGIOLYMPHATIC INVASION.**
B. **PATHOLOGIC STAGE: pT3 NX.**
C. **NODULAR THYROID HYPERPLASIA WITH SEPARATE ONCOCYTIC HYPERPLASTIC NODULE (1.8 CM).**
D. **INTRATHYMIC NORMOCELLULAR PARATHYROID TISSUE.**
E. **THYMUS WITH NO SIGNIFICANT ABNORMALITIES.**
F. **INCIDENTAL BENIGN SALIVARY CYST HETEROTOPIA COMPLEX.**

Criteria	Yes	No
Diagnosis Discrepancy		X
Primary Tumor Site Discrepancy		X
HIPAA Discrepancy		X
Prior Malignancy History		X
Dual/Synchronous Primary Noted		X
Case is (circle): QUALIFIED / DISQUALIFIED		
Reviewer Initials [illegible] Date Reviewed: [illegible]		